Advanced Mathematical Techniques First Day Warm-Up
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The purpose of this warm-up is to remind you about the character of infinite sequences, both those given explicitly and those given recursively. Along the way, I hope that you will get some idea of the need for a theorem in order to make definitive statements about the behavior of mathematical objects. You may use a calculator whenever you like. Give all answers either exactly or to at least 4 decimal places. Please show all of your work on a separate piece of paper.

1. Consider the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_n = \dfrac{(-1)^n n}{n+1}$.
   (a) Write out the first 5 terms of the sequence. What general behavior can you deduce from these terms?

   (b) It this sequence monotonic? Is it bounded? Would you say that the sequence converges or diverges? If it diverges, would you say that it contains a convergent subsequence?

   (c) Determine the limit point(s) of this sequence.

2. Consider the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_1 = 3$ and $a_{n+1} = 7 - 2/a_n$.
   (a) Write out the first 5 terms of the sequence. What general behavior of the terms do your results suggest? Does it appear to be monotonic? Does it appear to be bounded?

   (b) Show that this sequence is bounded, $2 < a_n < 7$ for all *n*, by induction (assume that $2 < a_k < 7$ for some value of *k*, then use the recurrence relation to show that $2 < a_{k+1} < 7$).

   (c) This sequence is indeed monotonic increasing. Prove this by induction. Your proof will be similar to that found in part (b).

   (d) Based on your analysis in (b) and (c), can you say for sure that this sequence converges? If so, what number (exactly) does it converge to?

3. Consider the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_1 = 3$ and $a_{n+1} = 7 + 2/a_n$.
   (a) Write out the first 5 terms of the sequence. What general behavior of the terms do your results suggest? Does it appear to be monotonic? Does it appear to be bounded?

   (b) This sequence is not monotonic, but it is bounded. Show using induction that the sequence is bounded by 2 from below and 12 from above. What convergence property can you state definitively from this result?

   (c) This sequence is convergent. Determine exactly what number it converges to and verify this result from investigating a few terms in the sequence.

4. Consider the sequence $\{S_n\}_{n=1}^{\infty}$, where $S_1 = 1$ and $S_{n+1} = S_n + \dfrac{1}{n+1}$.

   (a) Write out the first 5 terms of this sequence. Does it appear to be monotonic? Does it appear to be bounded?

   (b) The difference between adjacent terms in this sequence, $S_{n+1} - S_n = \dfrac{1}{n+1}$, clearly goes to zero as $n \to \infty$. Does this imply that the sequence is convergent? Why or why not?

   (c) Prove that this sequence is monotonic (it is not difficult, and does not require induction). Does this mean that it converges?

   (d) Work through the logic of the following expression:

$$S_n = 1 + \frac{1}{2} + \underbrace{\frac{1}{3} + \frac{1}{4}}_{\substack{>2\cdot 1/4 \\ <2\cdot 1/2}} + \underbrace{\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}}_{\substack{>4\cdot 1/8 \\ <4\cdot 1/4}} + \underbrace{\frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \frac{1}{15} + \frac{1}{16}}_{\substack{>8\cdot 1/16 \\ <8\cdot 1/8}} + \frac{1}{17} + \cdots + \frac{1}{n}$$

   Can you state conclusively whether or not this sequence converges based on this analysis? If it converges, can you give a bound that its limit is definitely less than?

5. Consider the sequence $\{S_n\}_{n=1}^{\infty}$, where $S_1 = 1$ and $S_{n+1} = S_n + \dfrac{1}{(n+1)^2}$.

   (a) Write out the first 5 terms of this sequence. Does it appear to be monotonic? Does it appear to be bounded?

   (b) The difference between adjacent terms in this sequence, $S_{n+1} - S_n = \dfrac{1}{(n+1)^2}$, clearly goes to zero as $n \to \infty$. Does this imply that the sequence is convergent? Why or why not?

   (c) Prove that this sequence is monotonic. Does this mean it converges?

   (d) Work through the logic of the following expression:

$$S_n = 1 + \frac{1}{2^2} + \underbrace{\frac{1}{3^2} + \frac{1}{4^2}}_{\substack{>2\cdot 1/4^2 \\ <2\cdot 1/2^2}} + \underbrace{\frac{1}{5^2} + \frac{1}{6^2} + \frac{1}{7^2} + \frac{1}{8^2}}_{\substack{>4\cdot 1/8^2 \\ <4\cdot 1/4^2}} + \underbrace{\frac{1}{9^2} + \frac{1}{10^2} + \frac{1}{11^2} + \frac{1}{12^2} + \frac{1}{13^2} + \frac{1}{14^2} + \frac{1}{15^2} + \frac{1}{16^2}}_{\substack{>8\cdot 1/16^2 \\ <8\cdot 1/8^2}} + \frac{1}{17^2} + \cdots + \frac{1}{n^2}$$

   Can you state conclusively whether or not this sequence converges based on this analysis? If it converges, can you give a bound that its limit is definitely less than?

6. Consider the sequence $\{a_n\}_{n=1}^{\infty}$, where $a_n = \cos\left(n^2\right)$.

   (a) Write out the first 5 terms of this sequence (decimal approximations are appropriate…). Does it appear to be monotonic? Does it appear to be bounded?

   (b) The difference between adjacent terms in this sequence, $a_{n+1} - a_n = \cos\left[\left(n+1\right)^2\right] - \cos\left(n^2\right)$, clearly does *not* go to 0 as $n \to \infty$. Does that mean the sequence does not converge?

   (c) Prove that this sequence is bounded. What does that imply about its convergence characteristics?

   (d) Can you prove that this sequence has at least one limit point? Can you find a limit point of this sequence? What is the difference between these two statements?